

O-Level Mathematics Performance Analysis

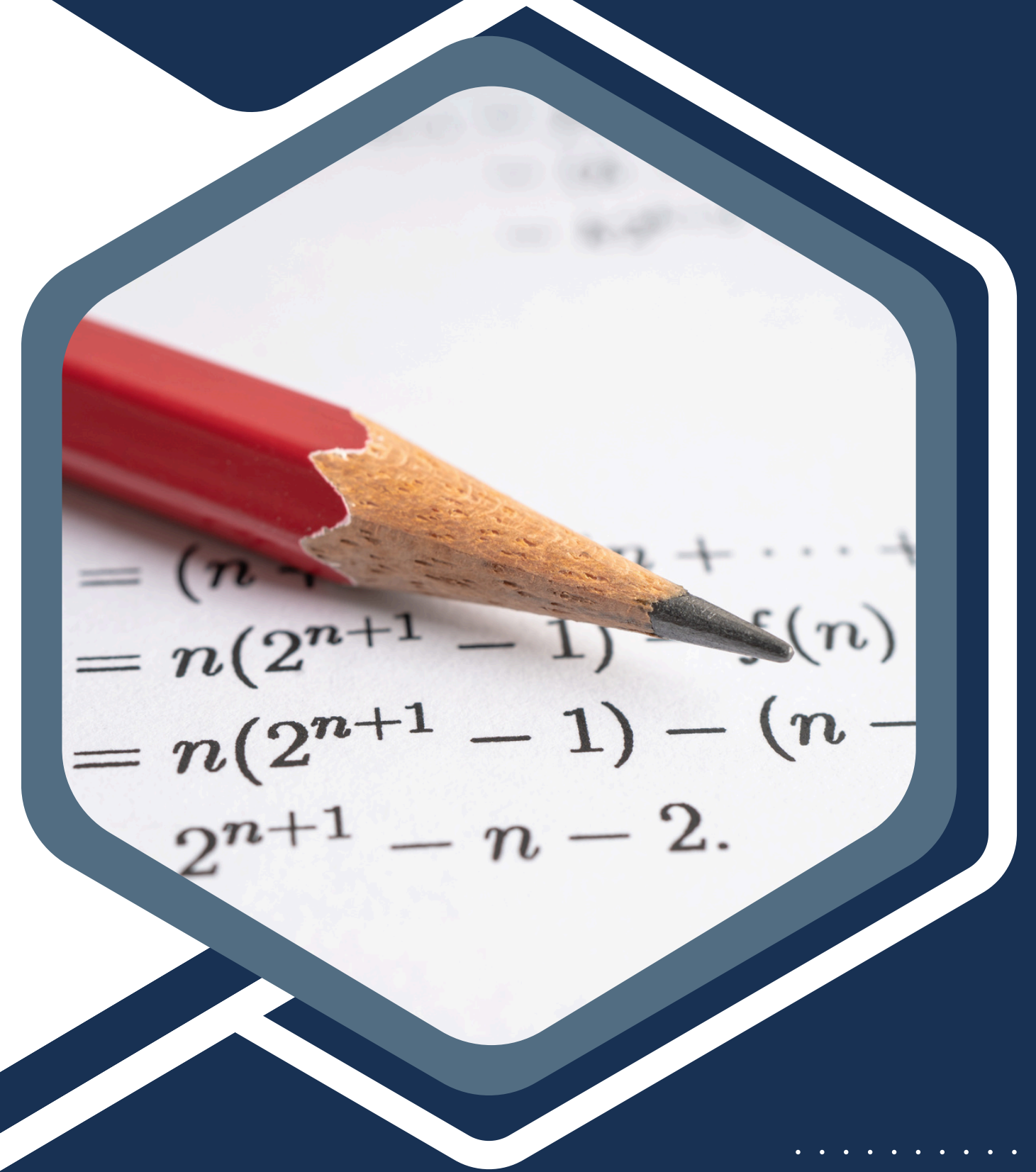
P r e s e n t a t i o n

PRESENTED BY:



MA EMILY SY

22 AUGUST 2026



Predicting O-Level Mathematics Performance for Early Academic Support

PROBLEM STATEMENT:

How can student academic habits, attendance and lifestyle characteristics be used to predict O-Level Mathematics performance and identify students who may benefit from early academic support?

OBJECTIVE:

- Predict a student's performance in Mathematics exam
- Identify potential risks
- Enable early human intervention



Who are our Key Stakeholders?



Ms. Tan Hui Min

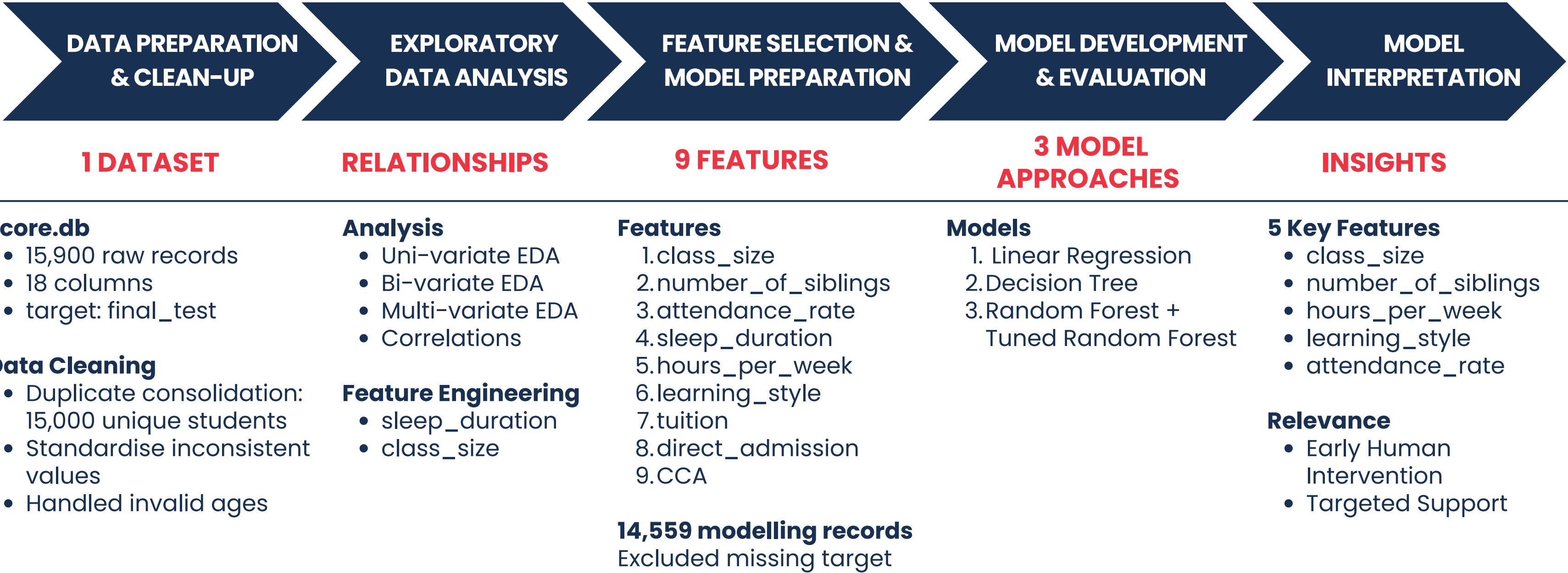
Senior Teacher, Academic Support
Meridian Crest Secondary School



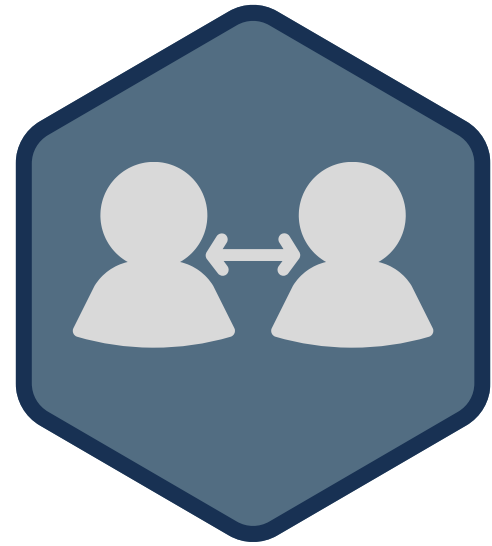
Mr Muhammad Irfan Rahman

Student Welfare Officer
Meridian Crest Secondary School

What's the Analytical Approach?



Data Cleaning before Building the Model



**Duplicate
Student IDs**



✓ **Consolidated** to
1 student record



**Complementary
missing values**



✓ **Recovered** available
attendance_rate & final_test



**Inconsistent
values**



✓ **Standardise** values
for CCA and tuition



**Invalid
ages**



✓ **Converted** invalid ages
outside valid 15–16 range
to missing

Feature Engineering



sleep_time & wake_time



Sleep Duration

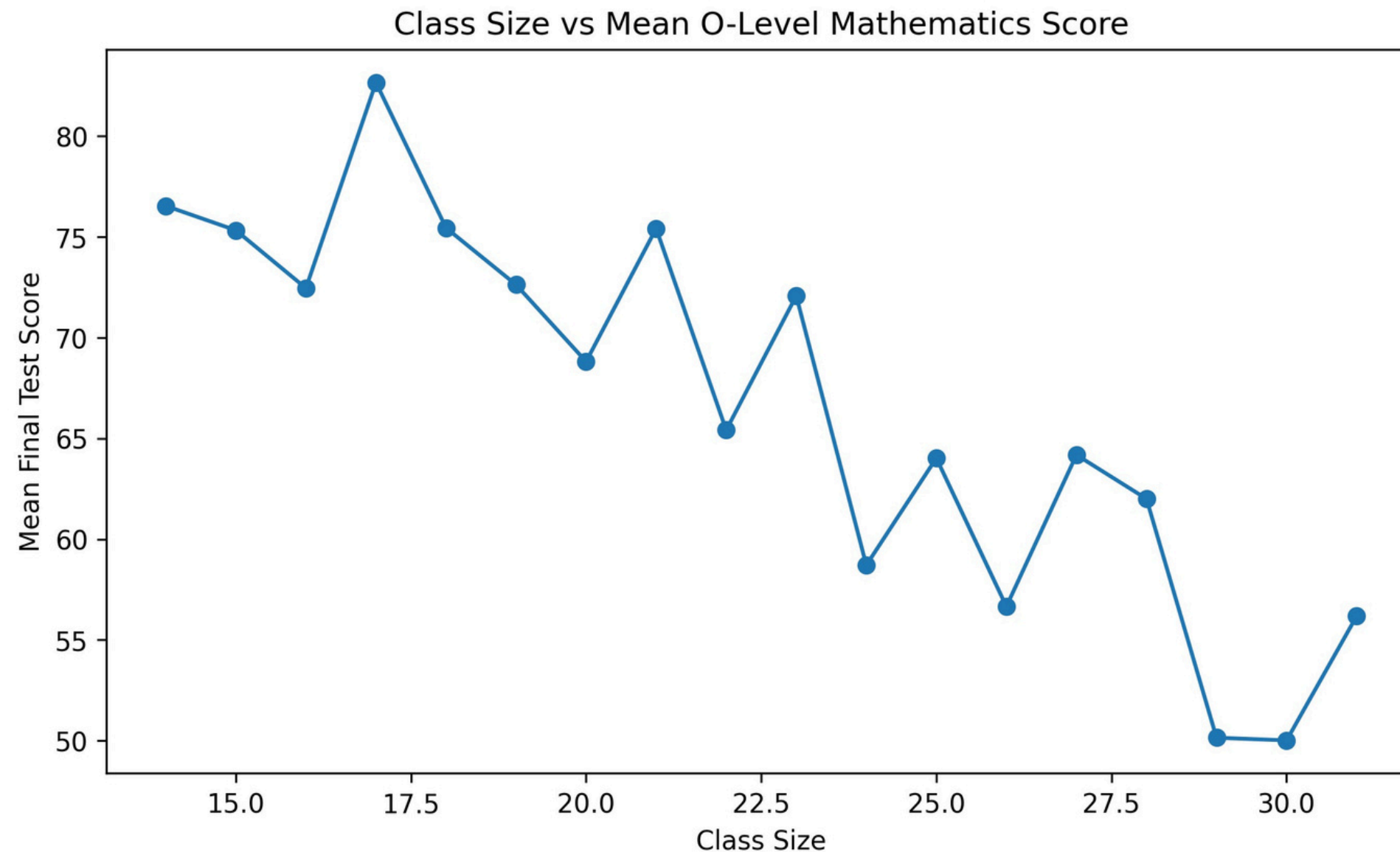


n_male + n_female



Class Size

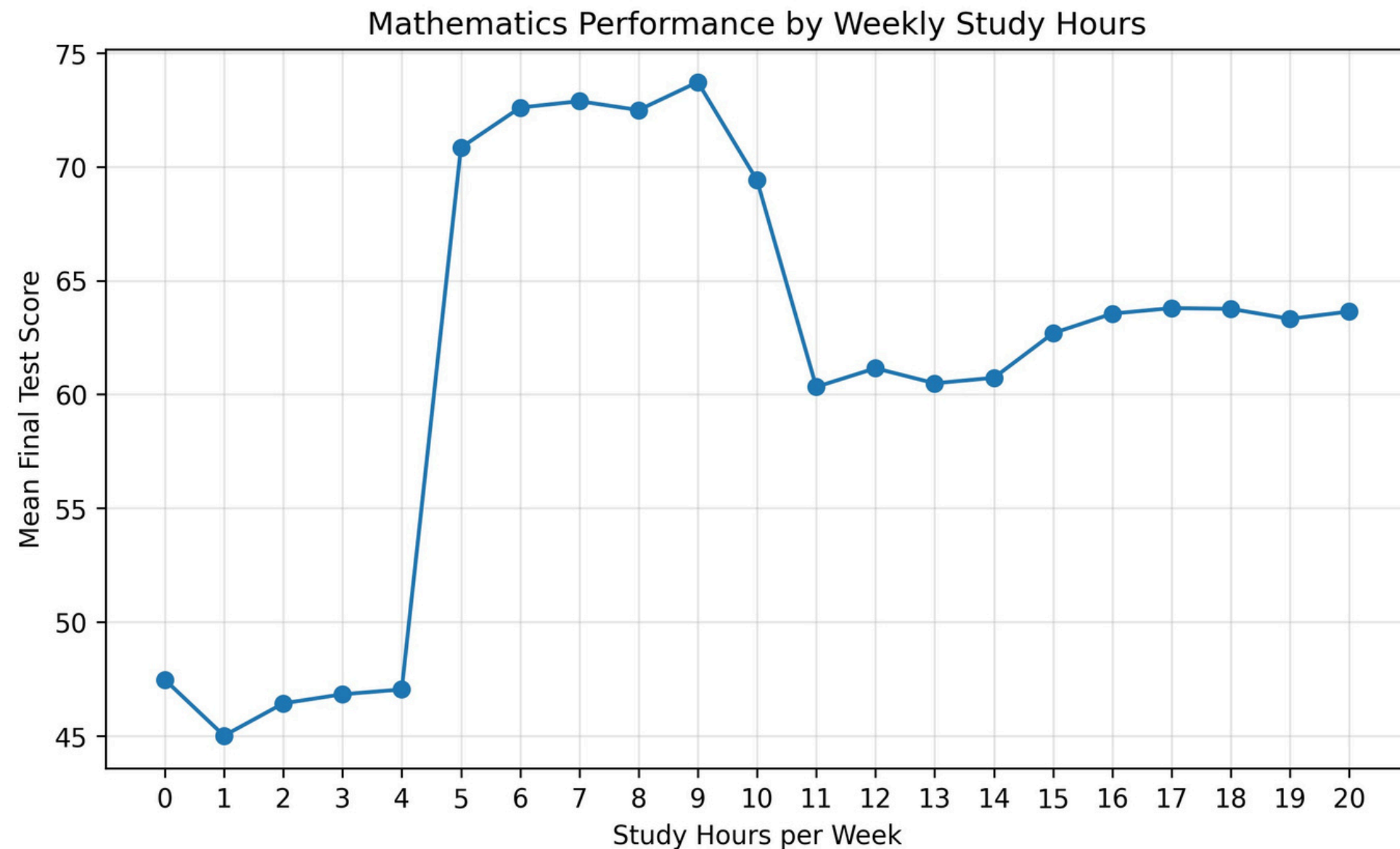
Smaller Classes Are Associated with Higher Mathematics Performance



Students in smaller classes generally achieved higher mean scores, while performance declined as class size increased.

Class size may therefore provide useful contextual information for identifying students who could benefit from additional academic support.

More Study Time Does Not Necessarily Mean Better Performance



Performance varies non-linearly with study time: students studying more hours did not consistently achieve higher scores.

Study hours should therefore be interpreted alongside other academic and lifestyle factors rather than as a standalone indicator.

Nine Predictors To Be Used For Modelling

Home Environment

01 number_of_siblings

Academic Behavior

02 attendance_rate

03 hours_per_week

Learning & Support

04 learning_style

05 tuition

06 direct_admission

Lifestyle

07 sleep_duration

School Environment

08 class_size

09 CCA

Random Forest: Strongest Baseline Model

Model	MAE	RMSE	Test R ²	R ² Gap	Remarks
Linear Regression	7.394	9.205	0.571	0.009	Stable, but unable to capture enough complexity
Decision Tree	4.825	7.316	0.729	0.254	Better prediction, but severe overfitting
Random Forest	4.007	5.925	0.822	0.143	Best unseen-data performance among the baseline models.

Random Forest Metric Evaluation:

- Average error $\approx$ 4.0 score points
- Larger errors remain, but are relatively limited
- Explains ~82% of variation in unseen test scores
- Noticeable overfitting

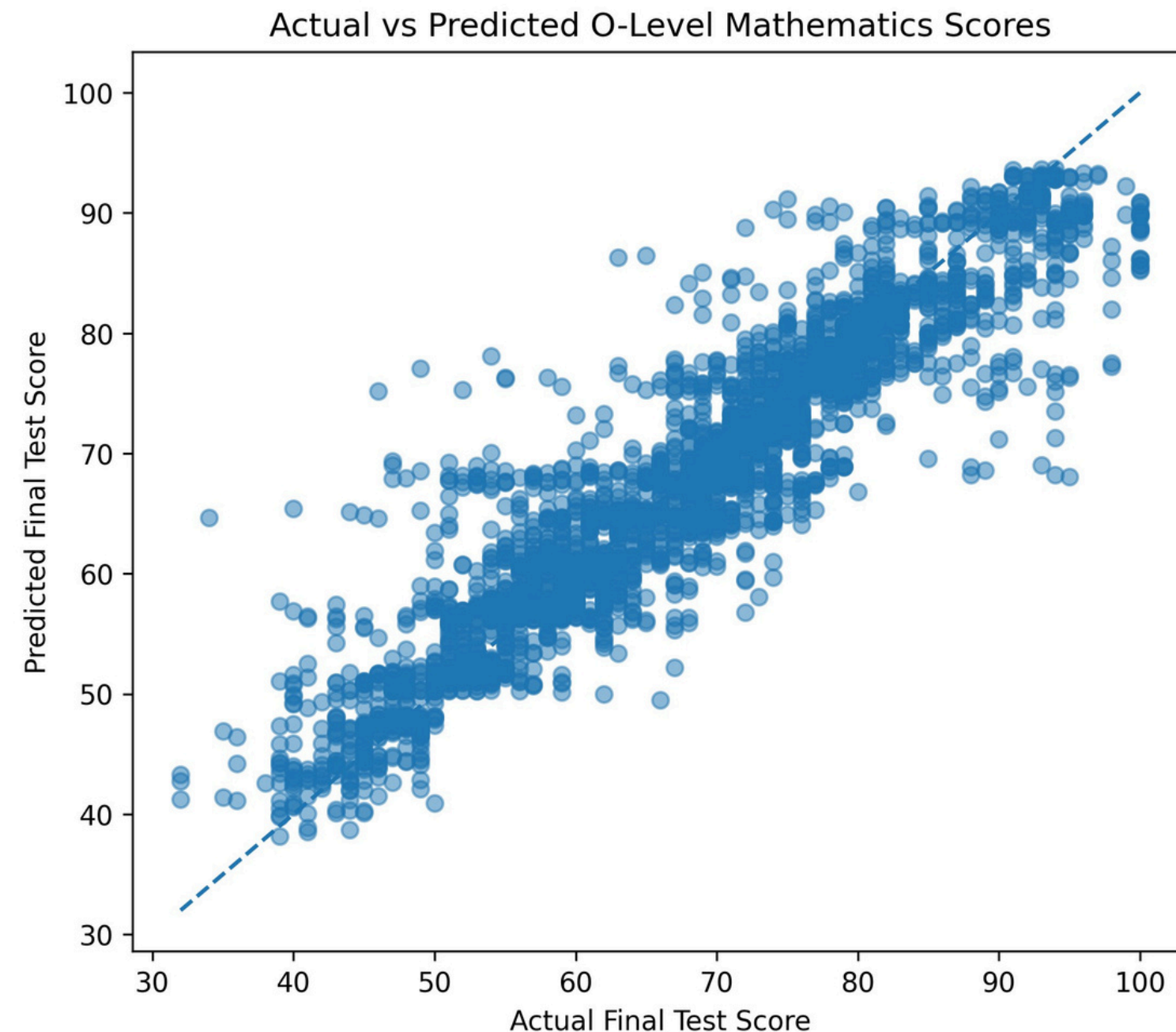
Tuned Random Forest: Better Performance, Less Overfitting

Metric	Score	Meaning
MAE	3.74	Average error $\approx$ 3.7 score points
RMSE	5.36	Larger errors remain, but are relatively limited
R ²	0.854	Explains ~85% of variation in unseen test scores
R ² Generalisation Gap	0.023	Small train-test gap indicates substantially improved generalisation From Baseline Random Forest: 0.143 Reduced by 0.120 (84%)

BEST PARAMETERS

384 trees • maximum depth=10 • max features=1.0 • minimum split=11 • minimum leaf size=2

Tuned Random Forest: Better Performance, Less Overfitting



Overall, predictions closely track actual Mathematics performance.

However, the model tends to moderate extreme scores—sometimes predicting low performers too high and very high performers too low.

This reinforces its use as an early screening tool, supported by human review.

Five Predictors Account for ~91% of Model Importance



Class Size

40.04%



**Number of
Siblings**

27.07%



**Study Hours
per week**

10.69%



**Learning
Style**

8.04%



**Attendance
Rate**

5.60%

Important Note: Predictive Importance $\neq$ Causation

From Prediction to Early Academic Support

Student Information

Gather

Academic Behavior, Home & School Environment, Learning & Support, and Lifestyle

ML Prediction

Generate

Predicted Mathematics Score

Risk Screening

Prioritize

Predicted Mathematics Score < Score Threshold

Human Review

Evaluate

Academic Support, Student Welfare & Teachers

Targeted Support

Act

Attendance Intervention • Study Support • Learning Support • Student Conversation

"Every child deserves a champion—an adult who will never give up on them, who understands the power of connection, and insists that they become the best that they can possibly be."

**Rita Pierson,
an educator for over 40 years**

Thank you!

**QUESTIONS, INSIGHTS, IDEAS?
LET'S CONTINUE THE CONVERSATION.**

www.linkedin.com/in/emilysy

APPENDIX

- **Data Cleaning Decisions**
- **EDA of Attendance Rate & Sleep Duration**
- **Feature Selection & EDA Evidence**
- **Model Evaluation**
- **Residual Analysis of Tuned Random Forest**
- **Class Size Band Analysis**
- **Pipeline Architecture**
- **Pipeline Hyperparameters**



Data Cleaning Decisions

Building a Reliable Modelling Dataset

Data Quality Issue

Decision

Duplicate Records

Consolidated duplicate student_id records, reducing 15,900 records to 15,000 unique students.

Missing Data

Removed 441 records with missing final scores; missing predictors such as attendance were handled through model preprocessing.

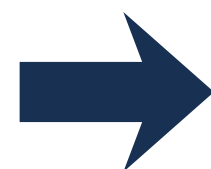
Invalid Age Values

Converted 451 invalid age values to missing rather than making assumptions. Age was later excluded from modelling.

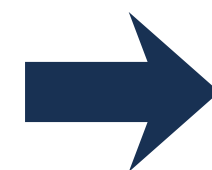
CCA Category Issue

Standardised inconsistent labels into Arts, Clubs, Sports and None to prevent duplicate categories.

15,900 Raw Records



15,000 Unique Students

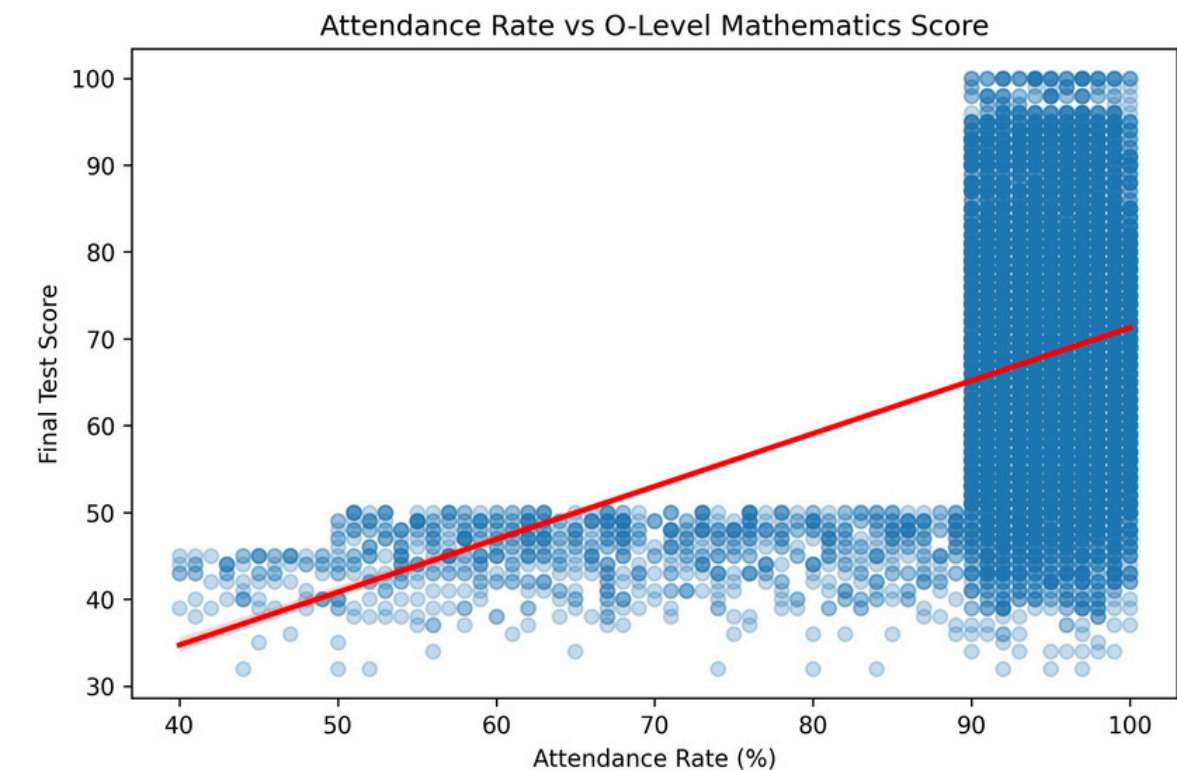


14,559 Modelling Records

Attendance and Sleep Reveal Different Student Patterns

1. Higher Attendance Is Generally Associated with Better Mathematics Performance

Students with higher attendance generally achieved stronger Mathematics outcomes, although substantial variation exists among students with similar attendance levels. Attendance provides useful predictive information, but should not be interpreted in isolation.



2. Short Sleep Coincides with a Broader At-Risk Student Profile

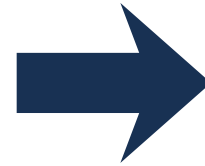
Students sleeping ≤ 6 hours showed:

- 59.65% average attendance
- 7.57 study hours/week
- 44.99 average Mathematics score

These relationships are associative and do not establish causality

Feature Selection & EDA Evidence

Numerical and categorical variables were evaluated based on:



- Distribution & data quality
- Relationship with final_test
- Meaningfulness to the problem statement
- Redundancy / limited predictive value

Area	EDA Evidence
Home Environment	• number_of_siblings – Scores varied across sibling counts, suggesting potential home-context influence
Academic Behavior	• hours_per_week – More study time did not consistently translate into higher scores • attendance_rate – Performance varied with attendance levels.
Learning & Support	• learning style – Score patterns differed between learning preferences. • direct admission – Performance differed by admission pathway. • tuition – Score patterns differed between students with and without tuition.
Lifestyle	• sleep_duration – Some variation in performance, but a relatively weak individual relationship
School Environment	• class_size – Smaller classes were associated with higher average scores • CCA – Some variation across CCA groups, but relatively modest differences

Model Evaluation

Tuned Random Forest delivered the best balance of predictive accuracy and generalisation. It achieved the **highest Test R^2 (0.856)** and **lowest Test MAE (3.72) and RMSE (5.34)**, while reducing the train–test R^2 gap to just **0.022**.

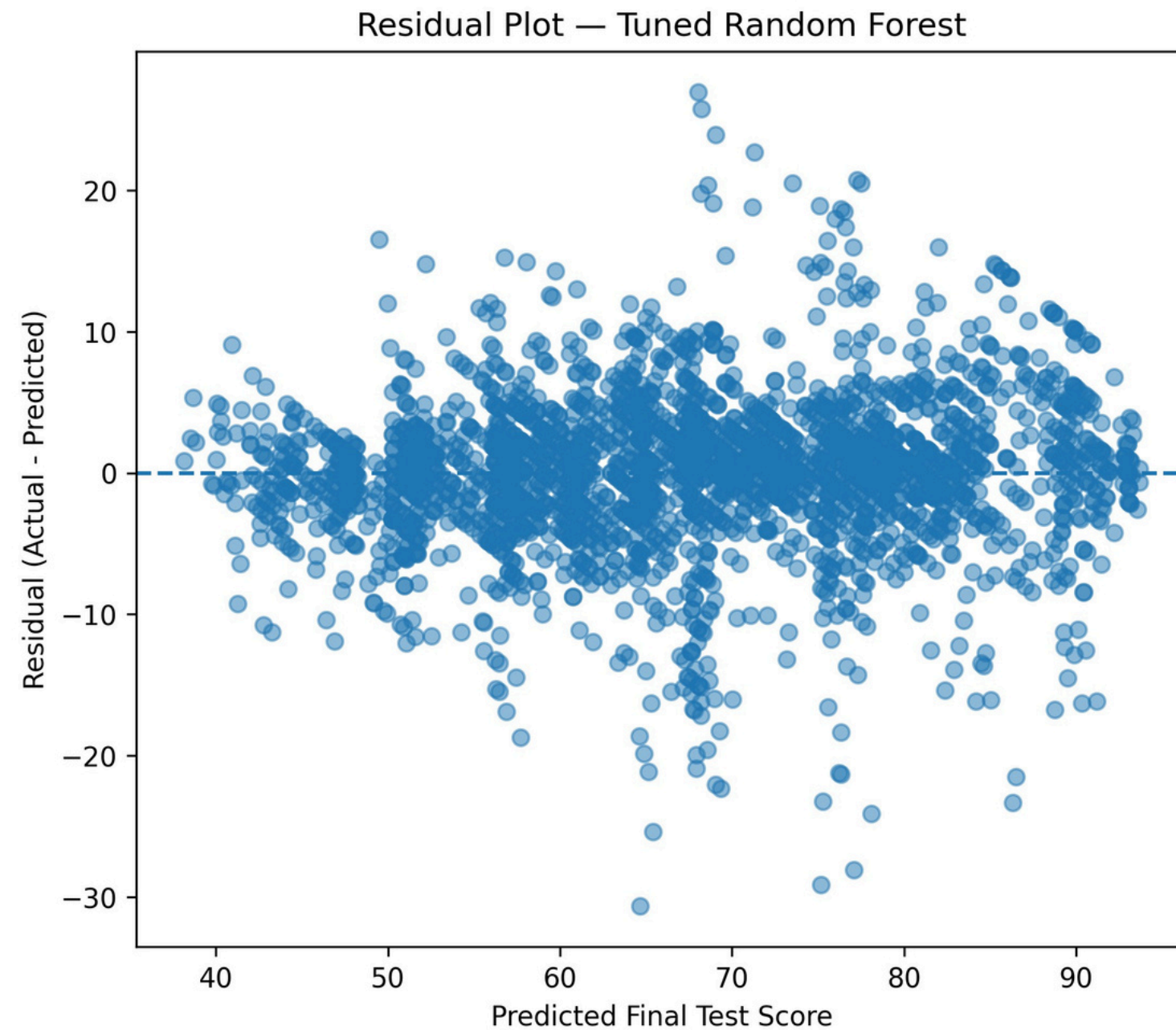
Model	Train MAE	Test MAE	Train RMSE	Test RMSE	Train R^2	Test R^2	R^2 Gap
Linear Regression	7.23	7.39	9.05	9.20	0.579	0.571	0.008
Decision Tree	0.58	4.81	1.87	7.31	0.982	0.729	0.253
Random Forest	1.68	4.00	2.62	5.92	0.965	0.822	0.143
Random Forest CV	1.65	3.92	2.55	5.73	0.966	0.831	0.135
Tuned Random Forest	3.36	3.72	4.88	5.34	0.878	0.856	0.022
Tuned Random Forest CV	3.35	3.74	4.84	5.36	0.88	0.852	0.027

5-fold cross-validation confirmed stable performance, with mean validation $R^2 = 0.852$, closely aligned with the independent test result.

Selected Model: Tuned Random Forest

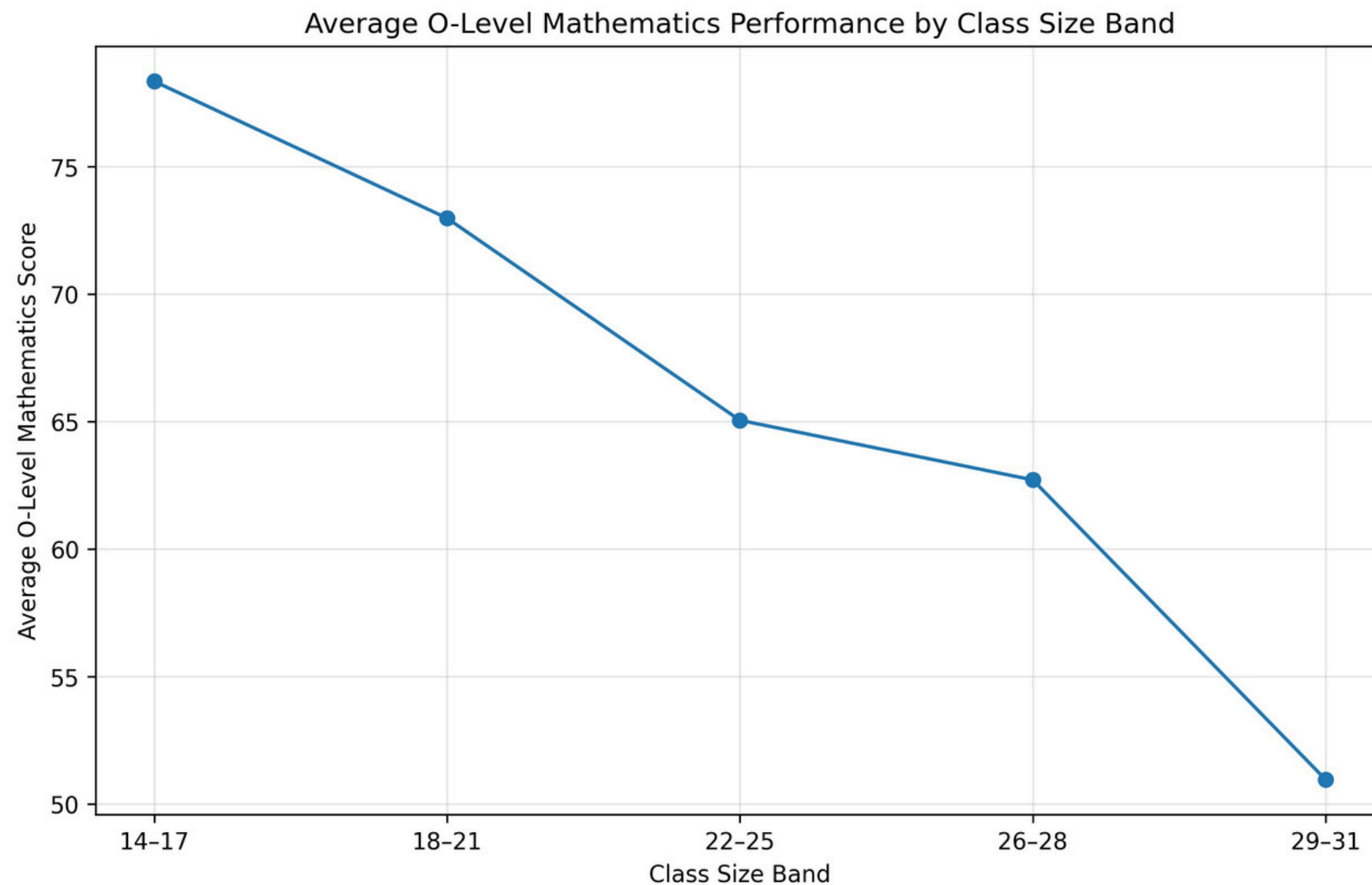
Explains approximately **86% of variation in Mathematics scores**, with **predictions differing from actual scores by 3.7 points** on average on the test set, while maintaining strong performance on unseen students.

Tuned Random Forest: Better Performance, Less Overfitting



- **76% of predictions are within ± 5 points of the actual score**
- 93% are within ± 10 points
- Larger prediction errors are uncommon: only 2.3% differ by more than 15 points.

Class Size Emerged as the Strongest Predictor of Mathematics Performance



Smaller classes are associated with stronger Mathematics performance.

14-17 students: 78%
18-21 students: 73%
22-25 students: 65%
26-28 students: 63%
29-31 students: 51%

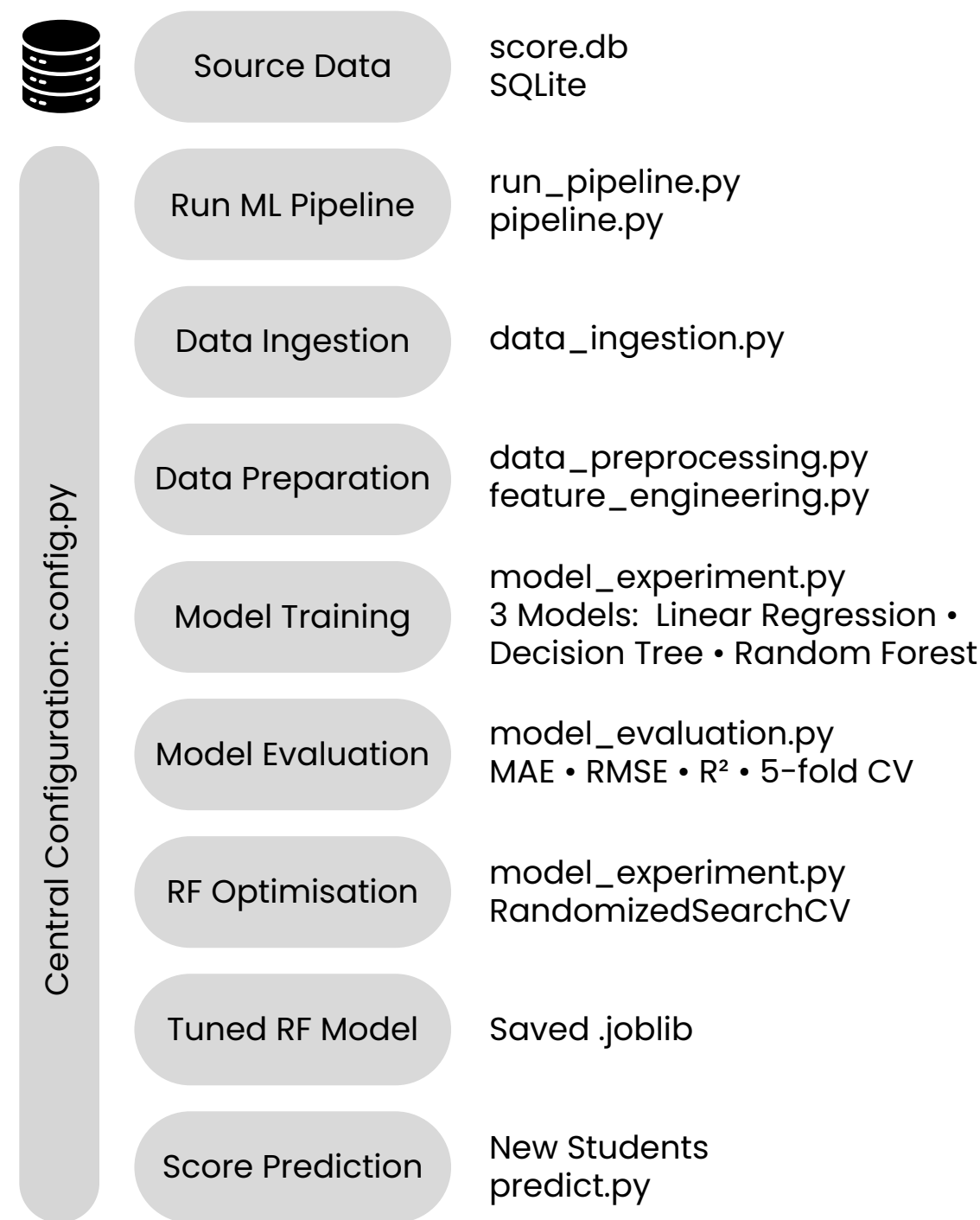
22+ students
emerges as a useful monitoring point

Consider class size alongside individual student risk indicators when prioritising students for early intervention.

Association, not causation. Class size should not be interpreted as independently causing differences in Mathematics performance.

Pipeline Architecture

End-to-End Pipeline Architecture



Pre-processing

Embedded in the scikit-learn pipeline to ensure the same transformations are applied during training and prediction.

Numerical Features

Median imputation for missing values

Categorical Features

One-Hot Encoding

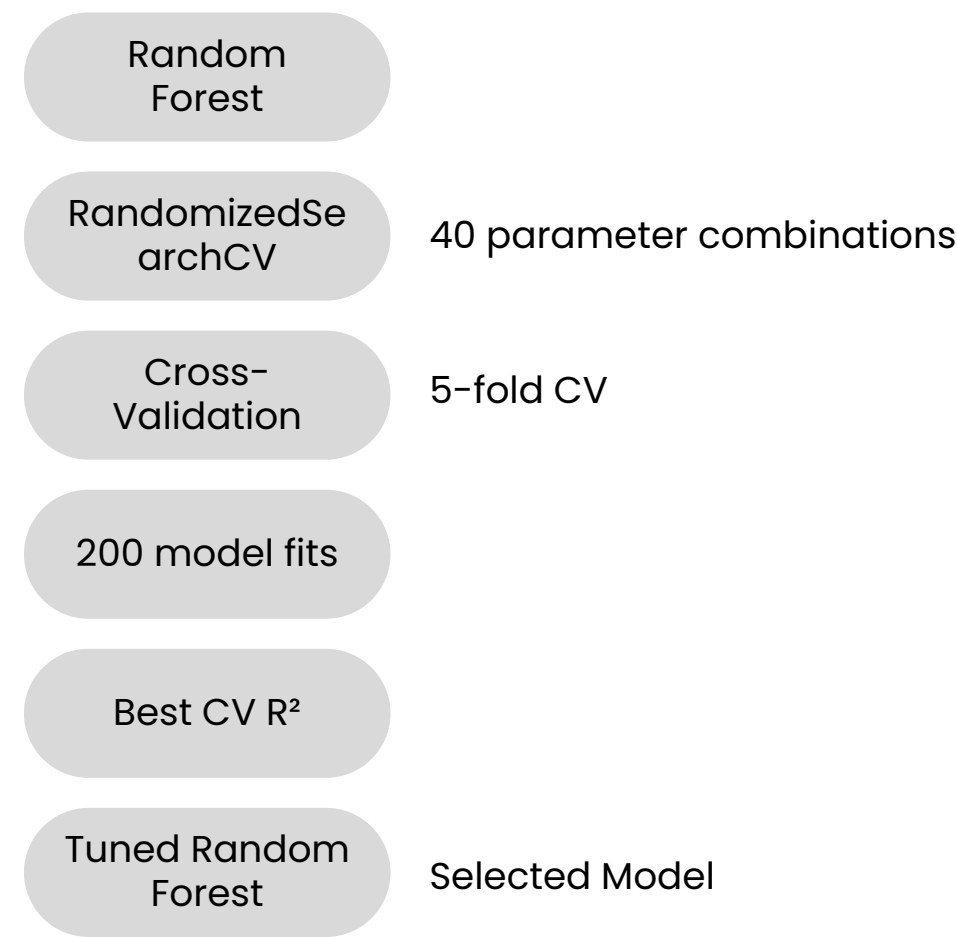
Feature Engineering

Sleep Duration & Class Size

Pipeline Hyperparameters

RandomizedSearchCV

Efficiently searches combinations of Random Forest hyperparameters using 5-fold cross-validation, without exhaustively testing every possible combination.



Random Forest Parameters

Hyperparameter	Selected Value	Meaning
n_estimators	384	Number of trees
max_depth	10	Maximum depth of each tree
max_features	1	All available features considered at each split
min_samples_split	11	Minimum samples required before splitting
min_samples_leaf	2	Minimum observations in a leaf